

Sequential Decision Making: Theory and Practice

Lecture 3: Markov Decision Processes

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Outline

Markov decision processes

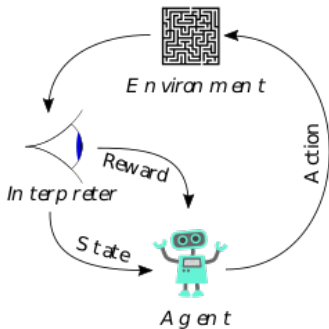
Policy evaluation

Policy improvement

Finding the optimal policy of MDPs

Reinforcement learning

In reinforcement learning (RL), an agent learns through interaction with the (unknown) environment.



RL has deep connection with control theory, and is also sometimes called **approximate dynamic programming**. It can be viewed as a type of **optimal control theory** with no pre-defined model of the environment.

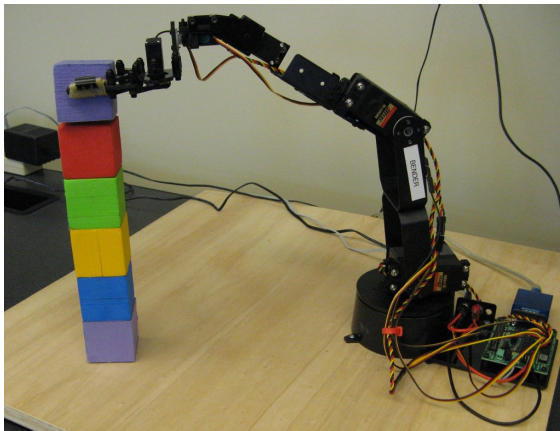
Applications of RL

RL can be applied to many different areas.

- **Robotics:** in which direction and how fast should a robot arm move?
- **Mobility:** where should taxis go to pick up passengers?
- **Transportation:** when should traffic lights turn green?
- **Recommendations:** which news stories will users click on?
- **Network configuration:** which parameter settings lead to the best allocation of resources?

Similar to multi-armed bandits, but with a notion of **state** or **context**.

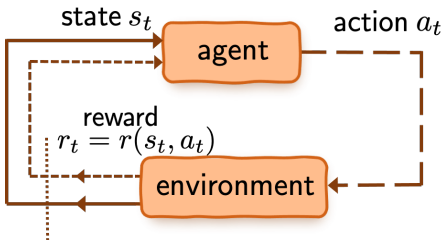
Example: grasping an object



RL **reinforces** the agents' decisions over time by observing the reward and state that result from taking different actions.

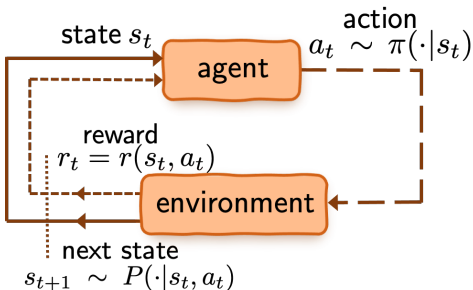
Markov decision processes

Infinite-horizon Markov decision process (MDP)



- $\mathcal{S}$: state space
- $\mathcal{A}$: action space
- $r(s, a) \in [0, 1]$: immediate reward

Infinite-horizon Markov decision process (MDP)



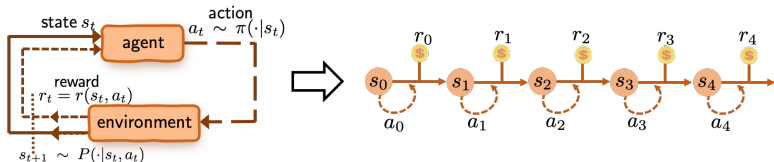
- $\mathcal{S}$: state space
- $\mathcal{A}$: action space
- $r(s, a) \in [0, 1]$: immediate reward
- $\pi(\cdot | s)$: policy (or action selection rule), deterministic or random
- $P(\cdot | s, a)$: transition probabilities

Help the mouse!



- state space $\mathcal{S}$: positions in the maze
- action space $\mathcal{A}$: up, down, left, right
- immediate reward r : cheese (+1), electricity shocks (-1), cats (-10000)
- policy $\pi(\cdot|s)$: the way to find cheese

Value function

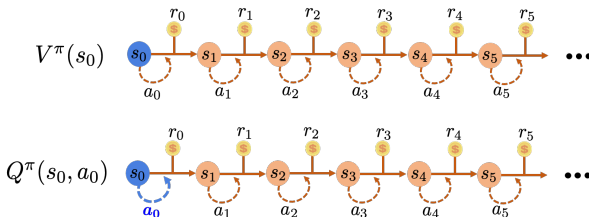


Value of policy π : cumulative **discounted** reward

$$\forall s \in \mathcal{S} : \quad V^\pi(s) := \mathbb{E} \left[\sum_{t=0}^{\infty} \gamma^t r(s_t, a_t) \mid s_0 = s \right]$$

- $(a_0, s_1, a_1, s_2, a_2, \dots)$: induced by policy π
- $\gamma \in [0, 1)$: discount factor,
 - γ close to 0 leads to **"myopic"** evaluation
 - γ close to 1 leads to **"far-sighted"** evaluation

Q-function (action-value function)



Q-function of policy π :

$$\forall (s, a) \in \mathcal{S} \times \mathcal{A}: \quad Q^\pi(s, a) := \mathbb{E} \left[\sum_{t=0}^{\infty} \gamma^t r_t \mid s_0 = s, a_0 = a \right]$$

- $(\cancel{a_0}, s_1, a_1, s_2, a_2, \dots)$: induced by policy π

Effective horizon

Since $r(s, a) \in [0, 1]$,

$$0 \leq V^\pi(s), Q^\pi(s, a) \leq \frac{1}{1 - \gamma}.$$

Often think of $\frac{1}{1 - \gamma}$ as the **effective horizon** of the problem.

Why Markov transitions?

- By the Markovian property,

$$\begin{aligned}P(s_{t+1}, s_t, \dots, s_0) &= P(s_0)P(s_1|s_0)P(s_2|s_1, s_0) \dots P(s_{t+1}|s_t, \dots, s_0) \\&= P(s_0)P(s_1|s_0)P(s_2|s_1, \cancel{s_0}) \dots P(s_{t+1}|s_t, \dots, \cancel{s_0}) \\&= P(s_0) \prod_{i=0}^t P(s_{i+1}|s_i).\end{aligned}$$

Low computation and memory complexity!

- The world is Markovian when the state space is large enough. For example, if $s_{t+1} \sim P(\cdot|s_t, s_{t-1})$ depends on the previous two steps, by working with $\tilde{s}_t = (s_t, s_{t-1})$ (and $s_{-1} = s_0$), we have

$$\tilde{s}_{t+1} \sim P(\cdot|\tilde{s}_t)$$

is Markovian.

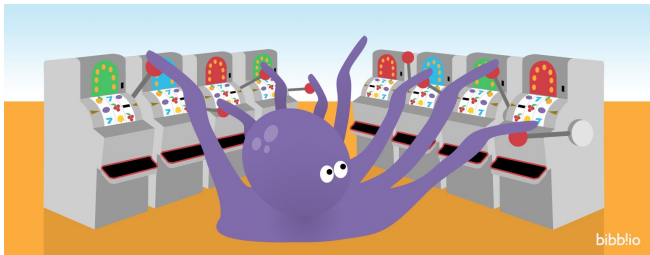
All models are wrong, but some are useful

Why discounting?



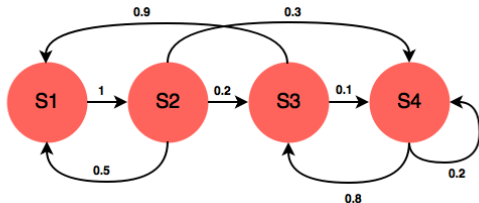
- Mathematically convenient: the limit always exists
- Immediate rewards earn more interest than future rewards
- Account for variability and uncertainty in the future which may not be fully captured
- Undiscounted MDP is possible, e.g. if all sequences terminate (like in a maze or game).
- Alternatives: average reward and finite-horizon episodic settings.

Reduction to multi-arm bandits



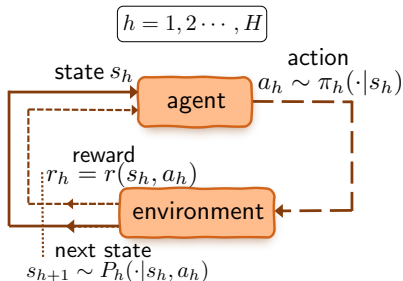
- No state transition: $\mathcal{S}$ is a singleton
- The reward function is action-dependent (action = arm): $r(a)$
- Short-horizon planning: discount factor $\gamma = 0$
- The value of a policy π becomes $V^\pi := \mathbb{E}_\pi[r(a)]$.

Reduction to Markov reward process



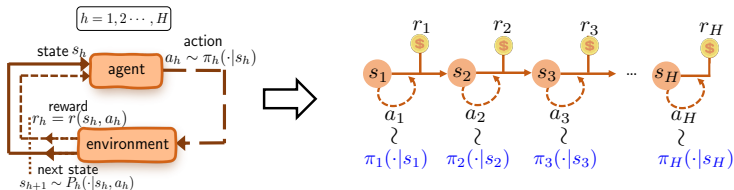
- No action selection: $\mathcal{A}$ is a singleton
- The transition kernel defines a Markov chain
- The reward function is state-dependent: $r(s)$
- The value becomes $V(s) := \mathbb{E}_{\pi}[\sum_{t=0}^{\infty} \gamma^t r(s_t) \mid s_0 = s]$.

Finite-horizon episodic MDP



- H : horizon length
- $\mathcal{S}$: state space with size S
- $\mathcal{A}$: action space with size A
- $r_h(s_h, a_h) \in [0, 1]$: immediate reward in step h
- $\pi = \{\pi_h\}_{h=1}^H$: policy (or action selection rule)
- $P_h(\cdot | s, a)$: transition probabilities in step h

Value function and Q-function



$$V_h^\pi(s) := \mathbb{E} \left[\sum_{t=h}^H r_t(s_t, a_t) \mid s_h = s \right]$$
$$Q_h^\pi(s, a) := \mathbb{E} \left[\sum_{t=h}^H r_t(s_t, a_t) \mid s_h = s, a_h = a \right]$$



- execute policy π to generate sample trajectory

Basic tasks

Policy evaluation:

- given a policy π , how good is it?

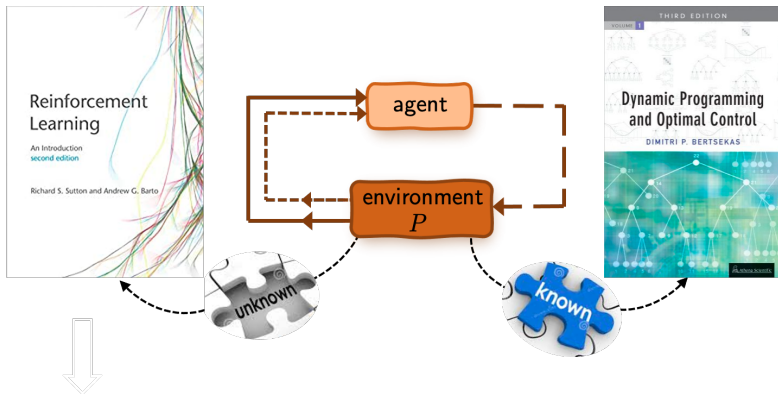
Policy improvements:

- given a policy π , can we find a better one?

Policy optimization:

- can we find the best policy for the given MDP?

Planning versus learning



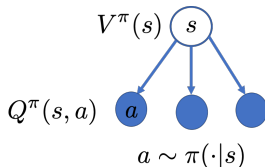
- **Planning:** solve for a desired policy given model specification
- **Learning:** learn a desired policy from samples w/o model specification

We'll focus on planning first.

Policy evaluation

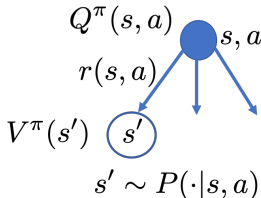
Policy evaluation: evaluating V via Q

$$\begin{aligned} V^\pi(s) &= \mathbb{E} \left[\sum_{t=0}^{\infty} \gamma^t r(s_t, a_t) \mid s_0 = s \right] \\ &= \sum_{a \in \mathcal{A}} \pi(a_0 = a \mid s = s_0) \underbrace{\mathbb{E} \left[\sum_{t=0}^{\infty} \gamma^t r(s_t, a_t) \mid s_0 = s, a_0 = a \right]}_{=: Q^\pi(s, a)} \\ &= \mathbb{E}_{a \sim \pi(\cdot \mid s)} [Q^\pi(s, a)] \end{aligned}$$



Policy evaluation: evaluating Q via V

$$\begin{aligned}Q^\pi(s, a) &= \mathbb{E} \left[r(s, a) + \sum_{t=1}^{\infty} \gamma^t r(s_t, a_t) \mid s_0 = s, a_0 = a \right] \\&= \mathbb{E} [r(s, a)] + \mathbb{E}_{s' \sim P(\cdot | s, a)} \left[\mathbb{E} \left[\sum_{t=1}^{\infty} \gamma^t r(s_t, a_t) \mid s_1 = s', s_0 = s, a_0 = a \right] \right] \\&= \mathbb{E} [r(s, a)] + \mathbb{E}_{s' \sim P(\cdot | s, a)} \left[\mathbb{E} \left[\sum_{t=1}^{\infty} \gamma^t r(s_t, a_t) \mid s_1 = s' \right] \right] \\&= \mathbb{E} [r(s, a)] + \underbrace{\gamma \mathbb{E}_{s' \sim P(\cdot | s, a)} \left[\mathbb{E} \left[\sum_{t=0}^{\infty} \gamma^t r(s_t, a_t) \mid s_0 = s' \right] \right]}_{=: V^\pi(s')} \\&= \mathbb{E} [r(s, a)] + \gamma \mathbb{E}_{s' \sim P(\cdot | s, a)} [V^\pi(s')]\end{aligned}$$



Bellman's consistency equation

- V^π / Q^π : value / action-value function under policy π

Bellman's consistency equation

$$V^\pi(s) = \mathbb{E}_{a \sim \pi(\cdot|s)} [Q^\pi(s, a)]$$
$$Q^\pi(s, a) = \underbrace{\mathbb{E}[r(s, a)]}_{\text{immediate reward}} + \gamma \mathbb{E}_{s' \sim P(\cdot|s, a)} \left[\underbrace{V^\pi(s')}_{\text{next state's value}} \right]$$

The value/Q function can be decomposed into two parts:

- immediate reward $\mathbb{E}[r(s, a)]$
- discounted value of at the successor state $\gamma \mathbb{E}_{s' \sim P(\cdot|s, a)} V(s')$



Richard Bellman

Matrix-vector representation

- Plugging Q^π into V^π , we have

$$V^\pi(s) = \mathbb{E}_{a \sim \pi(\cdot|s)}[r(s, a)] + \gamma \mathbb{E}_{a \sim \pi(\cdot|s), s' \sim P(\cdot|s, a)}[V^\pi(s')].$$

- Let P^π be the state-state transition matrix induced by π , namely,

$$P^\pi(s'|s) = \sum_{a \in \mathcal{A}} \pi(a|s) P(s'|s, a).$$

- We can write the above in a matrix-vector form as

$$V^\pi = r^\pi + \gamma P^\pi V^\pi,$$

where $V^\pi = [V^\pi(s)]_{s \in \mathcal{S}}$, and $r^\pi = [\mathbb{E}_{a \sim \pi(\cdot|s)}[r(s, a)]]_{s \in \mathcal{S}}$.

a similar treatment applies to Q^π , too.

Solving the Bellman's consistency equation

$$V^\pi = r^\pi + \gamma P^\pi V^\pi \implies V^\pi = (I - \gamma P^\pi)^{-1} r^\pi$$

Invertibility of $I - \gamma P^\pi$: Gershgorin's circle theorem, or for any $x \in \mathbb{R}^{|\mathcal{S}|}$, verify

$$\begin{aligned} \|(I - \gamma P^\pi)x\|_\infty &\geq \|x\|_\infty - \gamma \|P^\pi x\|_\infty \\ &\geq \|x\|_\infty - \gamma \|x\|_\infty \quad (\|P^\pi x\|_\infty \leq \|P^\pi\|_1 \|x\|_\infty = \|x\|_\infty) \\ &\geq (1 - \gamma) \|x\|_\infty \\ &> 0. \end{aligned}$$

Thus, $I - \gamma P^\pi$ is full rank and invertible.

Computationally expensive for problems with large state space!

Bellman's policy operator

Bellman's policy operator: denote the operator $\mathcal{T}^\pi : \mathbb{R}^{|S|} \mapsto \mathbb{R}^{|S|}$ as

$$\forall V \in \mathbb{R}^{|S|} : \quad \mathcal{T}^\pi(V) = r^\pi + \gamma P^\pi V.$$

Fixed-point equation:

$$V = \mathcal{T}^\pi(V)$$

- V^π is the unique fixed point of this fixed-point equation.

Contraction property of the Bellman's operator

Lemma 1

The operator $\mathcal{T}^\pi$ is a γ -contraction on $\mathbb{R}^{|\mathcal{S}|}$, i.e. for any V and V' in $\mathbb{R}^{|\mathcal{S}|}$, it follows

$$\|\mathcal{T}^\pi(V) - \mathcal{T}^\pi(V')\|_\infty \leq \gamma \|V - V'\|_\infty.$$

Proof: For any V and V' ,

$$\begin{aligned}\|\mathcal{T}^\pi(V) - \mathcal{T}^\pi(V')\|_\infty &= \|\gamma P^\pi V - \gamma P^\pi V'\|_\infty \\ &\leq \gamma \|P^\pi\|_1 \|V - V'\|_\infty = \gamma \|V - V'\|_\infty,\end{aligned}$$

using $\|P^\pi\|_1 = 1$.

Fast computation without inversion

Value iteration for policy evaluation

For $t = 0, 1, 2, \dots$

$$V^{(t+1)} = \mathcal{T}^\pi(V^{(t)}).$$

Linear convergence:

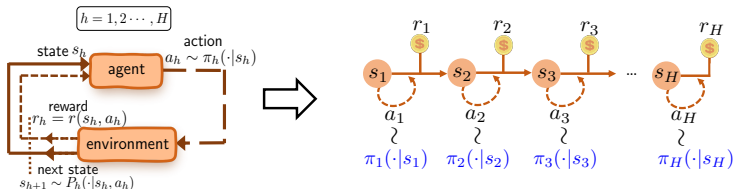
$$\begin{aligned}\|V^{(t+1)} - V^\pi\|_\infty &= \|\mathcal{T}^\pi(V^{(t)}) - \mathcal{T}^\pi(V^\pi)\|_\infty \\ &\leq \gamma \|V^{(t)} - V^\pi\|_\infty \\ &\leq \gamma^t \|V^{(0)} - V^\pi\|_\infty.\end{aligned}$$

Implication: to achieve $\|V^{(t+1)} - V^\pi\|_\infty \leq \epsilon$, it takes no more than

$$\frac{1}{1-\gamma} \log \left(\frac{\|V^{(0)} - V^\pi\|_\infty}{\epsilon} \right)$$

iterations.

Policy evaluation for finite-horizon MDPs



- 1 Begin with the terminal step $h = H + 1$:

$$V_{H+1}^\pi = 0, \quad Q_{H+1}^\pi = 0.$$

- 2 Backtrack $h = H, H - 1, \dots, 1$:

$$Q_h^\pi(s, a) := \underbrace{\mathbb{E}[r_h(s_h, a_h)]}_{\text{immediate reward}} + \underbrace{\mathbb{E}_{s' \sim P_h(\cdot | s, a)} V_{h+1}^\pi(s')}_{\text{next step's value}}$$

$$V_h^\pi(s) := \mathbb{E}_{a \sim \pi_h(\cdot | s)} Q_h^\pi(s, a)$$

Policy improvement

Partial ordering of policies

Definition 2 (Partial ordering)

Define a partial order over policies: denote

$$\pi' \geq \pi \quad \text{if} \quad \forall s \in \mathcal{S}, \quad V^{\pi'}(s) \geq V^{\pi}(s).$$

- The policy π' is an *improvement* over π since it improves its value in all states.

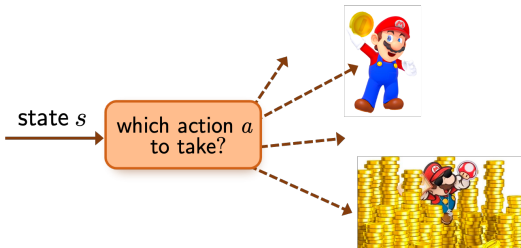
Question

Given a policy π , how to find an improved policy?



Policy improvement via one-step look-ahead

Given the Q-function Q^π of some policy π .



At each state s , can we identify an action a such that

$$Q^\pi(s, a) \geq V^\pi(s)?$$

- taking action a at state s leads to a higher cumulative reward than following policy π .

Policy improvement theorem

Theorem 3 (Policy improvement theorem)

Choose some stationary policy π , and let π' be a deterministic policy such that

$$\forall s \in \mathcal{S}, \quad Q^\pi(s, \pi'(s)) \geq V^\pi(s).$$

Then $V^{\pi'} \geq V^\pi$, i.e., π' is an improvement over π .

- Define the **greedy** policy w.r.t. some Q as

$$\pi_Q = \text{Greedy}(Q), \quad \text{i.e.} \quad \pi_Q(s) = \arg \max_{a \in \mathcal{A}} Q(s, a).$$

- The greedy policy $\pi' = \text{Greedy}(Q^\pi)$ w.r.t. Q^π is an improvement over π :

$$\begin{aligned} Q^\pi(s, \pi'(s)) &= \max_{a \in \mathcal{A}} Q^\pi(s, a) \\ &\geq \sum_{a \in \mathcal{A}} \pi(a|s) Q^\pi(s, a) = V^\pi(s) \quad \implies \quad V^{\pi'} \geq V^\pi \end{aligned}$$

Proof of policy improvement theorem

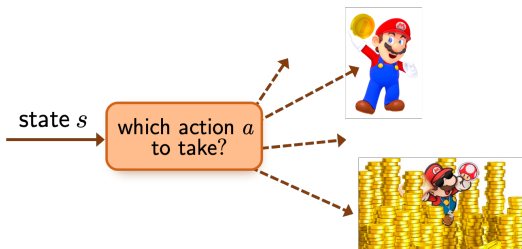
For each state $s \in \mathcal{S}$,

$$\begin{aligned} V^\pi(s) &\leq Q^\pi(s, \pi'(s)) \\ &= \mathbb{E} [r(s_0, \pi'(s_0)) + \gamma V^\pi(s_1) \mid s_0 = s, a_0 = \pi'(s)] \\ &\leq \mathbb{E} [r(s_0, \pi'(s_0)) + \gamma Q^\pi(s_1, \pi'(s_1)) \mid s_0 = s, a_0 = \pi'(s)] \\ &= \mathbb{E} [r(s_0, \pi'(s_0)) + \gamma r(s_1, \pi'(s_1)) + \gamma^2 V^\pi(s_2) \mid s_0 = s, a_0 = \pi'(s)] \\ &\leq \dots \\ &\leq \mathbb{E} [r(s_0, \pi'(s_0)) + \gamma r(s_1, \pi'(s_1)) + \gamma^2 r(s_2, \pi'(s_2)) + \dots \mid s_0 = s] \\ &\leq V^{\pi'}(s) \end{aligned}$$

One-step improvement leads to value increase.

Finding the optimal policy of MDPs

Optimal value and optimal policy



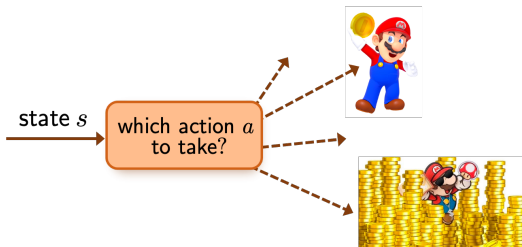
- **optimal value / Q function:**

$$V^*(s) := \max_{\pi} V^{\pi}(s), \quad Q^*(s, a) := \max_{\pi} Q^{\pi}(s, a)$$

where the search is over all policies possibly non-stationary and random.

- **optimal policy π^* :** the policy that maximizes the value function.

Optimal policy: existence



Lemma 4 ([Bellman, 1952])

For infinite-horizon discounted MDPs, there always exists a stationary and deterministic policy π^ , such that for all $s \in \mathcal{S}$, $a \in \mathcal{A}$,*

$$V^{\pi^*}(s) = V^*(s), \quad Q^{\pi^*}(s, a) = Q^*(s, a).$$

- Using stationary and deterministic policies suffices.
- See [Agarwal et al., 2019] for a proof.

Bellman's optimality equations

Theorem 5 (Bellman's optimality equations)

The optimal value/Q functions are unique and related via

$$\begin{aligned} V^*(s) &= \max_{a \in \mathcal{A}} Q^*(s, a), \\ Q^*(s, a) &= \mathbb{E}[r(s, a)] + \gamma \mathbb{E}_{s' \sim P(\cdot | s, a)} V^*(s'). \end{aligned}$$

Furthermore, $\pi^ = \pi_{Q^*} = \text{Greedy}(Q^*)$ is an optimal policy (tie-breaking arbitrarily).*

- Knowing the optimal Q-function allows us to find the optimal policy.
- The optimal values are unique, but the optimal policy is not necessarily unique.

Proof of Bellman's optimality equations

Proof of $V^*(s) = \max_{a \in \mathcal{A}} Q^*(s, a)$:

$$\begin{aligned} V^*(s) &= \max_{\pi} V^{\pi}(s) \\ &= \max_{\pi} \mathbb{E}_{a \sim \pi(\cdot|s)}[Q^{\pi}(s, a)] \\ &\leq \max_{\pi} \mathbb{E}_{a \sim \pi(\cdot|s)}[Q^*(s, a)] \\ &= \max_{a \in \mathcal{A}} Q^*(s, a). \end{aligned}$$

On the other end, for any $a \in \mathcal{A}$, consider the (possibly non-stationary) policy that first takes action a and then follows π^* . It follows that

$$V^*(s) \geq Q^{\pi^*}(s, a) = Q^*(s, a).$$

This implies, by the arbitrariness of a ,

$$V^*(s) \geq \max_{a \in \mathcal{A}} Q^*(s, a).$$

Proof of Bellman's optimality equations

Proof of $Q^*(s, a) = \mathbb{E}[r(s, a)] + \gamma \mathbb{E}_{s' \sim P(\cdot|s, a)} V^*(s')$:

$$\begin{aligned} Q^*(s, a) &= \max_{\pi} Q^{\pi}(s, a) \\ &= \max_{\pi} \left[\mathbb{E}[r(s, a)] + \gamma \mathbb{E}_{s' \sim P(\cdot|s, a)} V^{\pi}(s') \right] \\ &= \mathbb{E}[r(s, a)] + \gamma \max_{\pi} \mathbb{E}_{s' \sim P(\cdot|s, a)} V^{\pi}(s') \\ &= \mathbb{E}[r(s, a)] + \gamma \mathbb{E}_{s' \sim P(\cdot|s, a)} V^*(s') \end{aligned}$$

Bellman's optimality principle

Bellman operator

$$\mathcal{T}(Q)(s, a) := \underbrace{\mathbb{E}[r(s, a)]}_{\text{immediate reward}} + \gamma \mathbb{E}_{s' \sim P(\cdot | s, a)} \underbrace{\left[\max_{a' \in \mathcal{A}} Q(s', a') \right]}_{\text{next state's value}}$$

- one-step look-ahead

Bellman's optimality equation: Q^* is the *unique* fixed point to

$$\mathcal{T}(Q^*) = Q^*$$

Uniqueness is immediately implied by the γ -contraction on the next slide (verify!).



Richard Bellman

Contraction of the Bellman's operator

Lemma 6 (γ -contraction of Bellman operator)

For any Q and Q' , it holds

$$\|\mathcal{T}(Q) - \mathcal{T}(Q')\|_\infty \leq \gamma \|Q - Q'\|_\infty.$$

Proof: $\|\mathcal{T}(Q) - \mathcal{T}(Q')\|_\infty$

$$\begin{aligned} &= \gamma \max_{s,a} \left| \mathbb{E}_{s' \sim P(\cdot|s,a)} \left[\max_{a' \in \mathcal{A}} Q(s', a') \right] - \mathbb{E}_{s' \sim P(\cdot|s,a)} \left[\max_{a' \in \mathcal{A}} Q'(s', a') \right] \right| \\ &\leq \gamma \max_{s,a} \mathbb{E}_{s' \sim P(\cdot|s,a)} \left| \max_{a' \in \mathcal{A}} Q(s', a') - \max_{a' \in \mathcal{A}} Q'(s', a') \right| \\ &\leq \gamma \max_{s,a} \mathbb{E}_{s' \sim P(\cdot|s,a)} \max_{a' \in \mathcal{A}} |Q(s', a') - Q'(s', a')| \\ &\leq \gamma \max_{s', a'} |Q(s', a') - Q'(s', a')| = \gamma \|Q - Q'\|_\infty. \end{aligned}$$

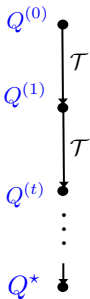
Here, we used the fact $|\max_a f(a) - \max_a g(a)| \leq \max_a |f(a) - g(a)|$.

Value iteration

Value iteration

For $t = 0, 1, \dots$,

$$Q^{(t+1)} = \mathcal{T}(Q^{(t)})$$



Convergence rate of value iteration

Theorem 7 (Linear convergence of value iteration)

$$\|Q^{(t)} - Q^*\|_\infty \leq \gamma^t \|Q^{(0)} - Q^*\|_\infty$$

- This is implied immediately by the γ -contraction property.

Implications: to achieve $\|Q^{(t)} - Q^*\|_\infty \leq \epsilon$, it takes no more than

$$\frac{1}{1 - \gamma} \log \left(\frac{\|Q^{(0)} - Q^*\|_\infty}{\epsilon} \right)$$

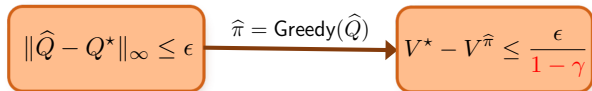
iterations.

From Q-function to policy

Lemma 8 ([Singh and Yee, 1994])

Let the greedy policy w.r.t. Q be π_Q , then

$$V^* - V^{\pi_Q} \leq \frac{2}{1-\gamma} \|Q^* - Q\|_\infty.$$



- Mind the error amplification factor $\frac{1}{1-\gamma}$

Proof of Lemma 8

Fix state $s \in \mathcal{S}$ and let $a = \pi_Q(s)$. It follows that

$$\begin{aligned} & V^*(s) - V^{\pi_Q}(s) \\ &= Q^*(s, \pi^*(s)) - Q^{\pi_Q}(s, \pi_Q(s)) \\ &= \underbrace{Q^*(s, \pi^*(s)) - Q(s, \pi_Q(s))}_{=: \text{I}} + \underbrace{Q(s, \pi_Q(s)) - Q^*(s, \pi_Q(s))}_{=: \text{II}} \\ &\quad + \underbrace{Q^*(s, \pi_Q(s)) - Q^{\pi_Q}(s, \pi_Q(s))}_{=: \text{III}} \end{aligned}$$

We shall bound each of these terms separately.

- For term I, since $Q(s, \pi_Q(s)) \geq Q(s, \pi^*(s))$,

$$\begin{aligned} Q^*(s, \pi^*(s)) - Q(s, \pi_Q(s)) &\leq Q^*(s, \pi^*(s)) - Q(s, \pi^*(s)) \\ &\leq \|Q^* - Q\|_{\infty}. \end{aligned}$$

Proof of Lemma 8

- For term II,

$$Q(s, \pi_Q(s)) - Q^*(s, \pi_Q(s)) \leq \|Q^* - Q\|_\infty.$$

- For term III, by Bellman equations,

$$\begin{aligned} Q^*(s, \pi_Q(s)) - Q^{\pi_Q}(s, \pi_Q(s)) &= \gamma \mathbb{E}_{s' \sim P(\cdot|s,a)} [V^*(s') - V^{\pi_Q}(s')] \\ &\leq \gamma \|V^* - V^{\pi_Q}\|_\infty \end{aligned}$$

To sum up,

$$\|V^* - V^{\pi_Q}\|_\infty \leq 2\|Q^* - Q\|_\infty + \gamma\|V^* - V^{\pi_Q}\|_\infty$$

$$\implies \|V^* - V^{\pi_Q}\|_\infty \leq \frac{2\|Q^* - Q\|_\infty}{1 - \gamma}$$

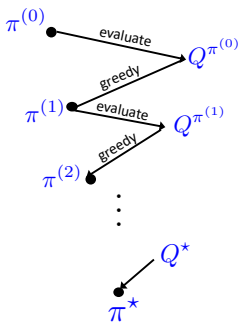
Policy iteration

Policy iteration

For $t = 0, 1, \dots$,

$$\pi^{(t)} = \text{Greedy}(Q^{(t-1)})$$

$$Q^{(t)} = Q^{\pi^{(t)}}$$



—“the dance”

Convergence rate of policy iteration

Theorem 9 (Linear convergence of policy iteration)

For policy iteration, it follows that

- 1 $Q^{(t+1)} \geq \mathcal{T}(Q^{(t)}) \geq Q^{(t)}$
- 2 $\|Q^{(t+1)} - Q^*\|_\infty \leq \gamma \|Q^{(t)} - Q^*\|_\infty$

- Policy iteration produces a sequence of improving policies.

Implications: to achieve $\|Q^{(t)} - Q^*\|_\infty \leq \epsilon$ for output policy $\pi^{(t)}$, it takes no more than

$$\frac{1}{1 - \gamma} \log \left(\frac{\|Q^{(0)} - Q^*\|_\infty}{\epsilon} \right)$$

iterations.

Proof for policy iteration: policy improvements

Proof of $\mathcal{T}(Q^{(t)}) \geq Q^{(t)}$:

$$\begin{aligned}\mathcal{T}(Q^{(t)})(s, a) &= r(s, a) + \gamma \mathbb{E}_{s' \sim P(\cdot|s, a)} \max_{a'} Q^{\pi^{(t)}}(s', a') \\ &\geq r(s, a) + \gamma \mathbb{E}_{s' \sim P(\cdot|s, a)} Q^{\pi^{(t)}}(s', \pi^{(t)}(s')) \\ &= r(s, a) + \gamma \mathbb{E}_{s' \sim P(\cdot|s, a)} V^{\pi^{(t)}}(s') = Q^{\pi^{(t)}}(s, a).\end{aligned}$$

Proof of $Q^{(t+1)} \geq \mathcal{T}(Q^{(t)})$: From policy improvement theorem, we already know $Q^{(t+1)} \geq Q^{(t)}$.

$$\begin{aligned}Q^{\pi^{(t+1)}}(s, a) &= r(s, a) + \gamma \mathbb{E}_{s' \sim P(\cdot|s, a)} Q^{\pi^{(t+1)}}(s', \pi^{(t+1)}(s')) \\ &\geq r(s, a) + \gamma \mathbb{E}_{s' \sim P(\cdot|s, a)} Q^{\pi^{(t)}}(s', \pi^{(t+1)}(s')) \\ &= r(s, a) + \gamma \mathbb{E}_{s' \sim P(\cdot|s, a)} \max_{a'} Q^{\pi^{(t)}}(s', a') = \mathcal{T}(Q^{(t)})(s, a).\end{aligned}$$

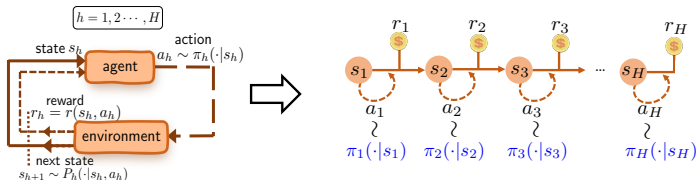
Proof for policy iteration: linear convergence

Using $Q^{(t+1)} \geq \mathcal{T}(Q^{(t)})$,

$$\begin{aligned}\|Q^* - Q^{(t+1)}\|_\infty &\leq \|Q^* - \mathcal{T}(Q^{(t)})\|_\infty \\ &= \|\mathcal{T}(Q^*) - \mathcal{T}(Q^{(t)})\|_\infty \\ &\leq \gamma \|Q^* - Q^{(t)}\|_\infty.\end{aligned}$$

Here, the last line follows from the contraction of the Bellman's optimality operator.

Bellman's optimality eq. for finite-horizon MDPs



Let $Q_h^*(s, a) = \max_{\pi} Q_h^{\pi}(s, a)$ and $V_h^*(s) = \max_{\pi} V_h^{\pi}(s)$.

- 1 Begin with the terminal step $h = H + 1$:

$$V_{H+1}^* = 0, \quad Q_{H+1}^* = 0.$$

- 2 Backtrack $h = H, H - 1, \dots, 1$:

$$Q_h^*(s, a) := \underbrace{\mathbb{E}[r_h(s_h, a_h)]}_{\text{immediate reward}} + \underbrace{\mathbb{E}_{s' \sim P_h(\cdot|s, a)} V_{h+1}^*(s')}_{\text{next step's value}}$$

$$V_h^*(s) := \max_{a \in \mathcal{A}} Q_h^*(s, a), \quad \pi_h^*(s) = \operatorname{argmax}_{a \in \mathcal{A}} Q_h^*(s, a).$$

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